

Generation Interconnection System Impact Study Report

For Queue Project Q22-2190

Ely 138 kV
November 15, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2190. The Interconnection Customer proposes a 82.0 MW solar photovoltaic facility interconnecting at the Ely 138 kV substation in the Virginia area. The request was received on September 26, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 683 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,473,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Ely – Wheelabrator South Broward 138 KV, the Ely – Nottoway Diesel Generating Facility 138 KV loads to 106.6% of its 222 MVA rating. Project DFAX: 100.0%.

For the loss of Nottoway Diesel Generating Facility – Ely 138 KV, the Ely – Wheelabrator S Broward 138 KV loads to 104.2% of its 227 MVA rating. Project DFAX: 100.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ely – Nottoway Diesel Generating Facility 138 KV	Reconductor line	\$5,262,000	33% (\$1,723,000)
Ely – Wheelabrator S Broward 138 KV	Reconductor line	\$5,346,000	33% (\$1,750,000)
	Total allocated to project		\$3,473,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.